PS-6

Arijit Das

2026-03-21

To demonstrate the application of subset selection methods

Consider the "Hitters" data. For loading the data you need to install ISLR library. We wish to predict a baseball player's salary on the basis of various predictors. So first we need to choose an optimal model.

```
rm(list=ls())
library("ISLR")
```

1. Clean the data by removing all rows that have missing value in any variable. You will be working with this clean data.

```
df=na.omit(Hitters)
dim(df)

## [1] 263  20
```

The cleaned Hitters dataset contains 263 observations and 19 predictor variables. Let Y_i denote the response variable (Salary) for the ith player and let X_{ij} denote the jth predictor for the ith player, where $j = 1,2, \dots ,19$.

The multiple linear regression model is given by

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \cdots + \beta_{19} X_{i19} + \varepsilon_i, \quad i = 1,2, \dots ,263$$

where

$$\varepsilon_i \sim N(0, \sigma^2)$$

Here,

- Y_i : Salary of the ith player
- X_{ij} : j^{th} predictor variable for the ith player
- β_0 : Intercept
- β_j : Regression coefficient (unknown) corresponding to the jth predictor for $j = 1,2, \dots ,19$
- ε_i : Random error term

i=1(1)263 and j=1(1)19

2. **Split the data randomly into train-test keeping 80% as training sample and remaining as test sample.**

```r
set.seed(123)
train.id=sample(1:263,210)
train.data=df[train.id,]
test.data=df[-train.id,]
dim(train.data)

## [1] 210  20


dim(test.data)

## [1] 53 20
```

Among the 263 rows we choose 80% rows i.e. 210 rows as the Training Data and the rest 53 rows as the Test data.

3. **Perform the best subset selection on the training sample. While choosing the best model fit up to a 19 variable model. Choose the best model according to adjusted R squared, BIC , AIC and Cp values. In case the best model differs according to the selection criteria, report the best fitted model according to each selection criterion.**

```r
library(leaps)

best.subset=regsubsets(Salary~.,data=train.data,nvmax=19)
adjr2=summary(best.subset)$adjr2
cp=summary(best.subset)$cp
bic=summary(best.subset)$bic
```

- The less the BIC, better is the model. So the fitted model corresponding to the lowest BIC value is given by,

```r
coef(best.subset, which.min(bic))

##  (Intercept)        Walks       CAtBat         CHits        CHmRun
DivisionW
##  149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -
129.0165012
```

$$\widehat{\text{Salary}} = 149.5854 + 5.9582 \cdot \text{Walks} - 0.5242 \cdot \text{CAtBat} + 2.0687 \cdot \text{CHits} + 1.7653 \cdot \text{CHmRun} - 129.0165 \cdot \text{DivisionW}$$

* The less the Cp value, better is the model. So the fitted model corresponding to the lowest Cp value is given by,

```r
coef(best.subset, which.min(cp))

##  (Intercept)        AtBat         Hits        Walks       CAtBat        CRuns
##  134.1593585   -1.9328699    5.8271814    6.5037245   -0.2063323    1.7820180
##         CRBI       CWalks    DivisionW      PutOuts      Assists   NewLeagueN
##    1.1120229   -0.8879140 -1 21.2135304    0.1925628    0.3316973   71.1433129
```

$$\widehat{\text{Salary}} = 134.1594 - 1.9329\text{AtBat} + 5.8272\text{Hits} + 6.5037\text{Walks} - 0.2063\text{CAtBat} + 1.7820\text{CRuns} + 1.1120\text{CRBI} - 0.8879\text{CWalks} - 121.2135\text{DivisionW} + 0.1926\text{PutOuts} + 0.3317\text{Assists} + 71.1433\text{NewLeagueN}$$

The higher the Adjusted R squared value, better is the model. So the fitted model corresponding to the highest adjusted R square value is given by,

```
coef(best.subset, which.max(adjr2)) #Best Adjusted R squared Model

##  (Intercept)        AtBat         Hits        Walks       CAtBat
CRuns
##  154.1594365   -1.8722264    5.7547106    6.3852096   -0.2092154
1.7712871
##         CRBI       CWalks    DivisionW      PutOuts      Assists
Errors
##    1.1419024   -0.8989348 -121.8633686    0.2013298    0.5363804   -
6.4726568
##   NewLeagueN
##   75.8378982
```

$$\widehat{\text{Salary}} = 154.1594 - 1.8722\text{AtBat} + 5.7547\text{Hits} + 6.3852\text{Walks} - 0.2092\text{CAtBat} + 1.7713\text{CRuns} + 1.1419\text{CRBI} - 0.8989\text{CWalks} - 121.8634\text{DivisionW} + 0.2013\text{PutOuts} + 0.5364\text{Assists} - 6.4727\text{Errors} + 75.8379\text{NewLeagueN}$$

4. Find the training MSE and test MSE for each of the best models selected according to different criterion as mentioned in 3 and compare their values.

```
#Train MSE and Test MSE
m.cp=lm(Salary~AtBat+Hits+Walks+CAtBat+CRuns+CRBI+CWalks+Division+PutOuts+Ass
ists+NewLeague,data=train.data)
m.bic=lm(Salary~Walks+CAtBat+CHits+CHmRun+Division,data=train.data)
m.adjr=lm(Salary~AtBat+Hits+Walks+CAtBat+CRuns+CRBI+CWalks+Division+PutOuts+A
ssists+Errors+NewLeague,data=train.data)
mse.train.cp=1/210*(sum((train.data$Salary-predict(m.cp))^2))
mse.train.bic=1/210*(sum((train.data$Salary-predict(m.bic))^2))
mse.train.adjr=1/210*(sum((train.data$Salary-predict(m.adjr))^2))
y.hat.test.cp=predict(m.cp, newdata=test.data)
y.hat.test.bic=predict(m.bic, newdata=test.data)
y.hat.test.adjr=predict(m.adjr, newdata=test.data)
mse.test.cp=mean((test.data$Salary-y.hat.test.cp)^2)
mse.test.bic=mean((test.data$Salary-y.hat.test.bic)^2)
mse.test.adjr=mean((test.data$Salary-y.hat.test.adjr)^2)
out.best=data.frame(Measure=c("Cp","BIC","AdjR2"),Training_MSE=c(mse.train.cp
,mse.train.bic,mse.train.adjr),
                    Test_MSE=c(mse.test.cp,mse.test.bic,mse.test.adjr))
out.best

##   Measure Training_MSE Test_MSE
## 1      Cp     89790.63 125652.3
```

```
## 2     BIC     97026.43 153090.5
## 3   AdjR2     88935.80 130066.9
```

Comment:

The results compare three models selected using different criteria in the best subset selection method: Cp, BIC, and Adjusted R^2.

Among these, the Cp-based model has the lowest test mean squared error (125652.3), indicating that it performs best on unseen data. The Adjusted R^2 model has a slightly higher test MSE (130066.9), suggesting reasonably good performance but not as strong as the Cp model. On the other hand, the BIC-based model has the highest test MSE (153090.5), indicating poorer predictive performance.

This suggests that the Cp criterion provides the best balance between model complexity and goodness of fit in this case, leading to better generalization. The Adjusted R^2 model performs moderately well, while the BIC model appears to underfit the data due to its stronger penalty on model complexity.

Therefore, for prediction purposes, the Cp-based model is preferred.

5. Repeat 3-4 using forward stepwise selection method.

```
forward.subset=regsubsets(Salary~.,data=train.data,nvmax=19,method="forward")
coef(forward.subset, which.min(summary(forward.subset)$bic)) #Best BIC Model

##  (Intercept)        Walks       CAtBat        CHits        CRuns         CRBI
##  127.8159745    6.4746441   -0.4127407    1.1133040    1.0341530    0.8520825
##       CWalks    DivisionW
##   -0.5432536 -127.3877839

coef(forward.subset, which.min(summary(forward.subset)$cp)) #Best cp Model

##  (Intercept)        Walks       CAtBat        CHits        CRuns         CRBI
##  110.1015291    5.2753942   -0.4498891    1.0510348    1.3425171    0.9381944
##       CWalks    DivisionW      PutOuts      Assists       Errors   NewLeagueN
##   -0.5426157 -125.5159872    0.1609710    0.4662319   -7.6994411   74.6774479

coef(forward.subset, which.max(summary(forward.subset)$adjr2)) #Best Adjusted R
squared Model

##  (Intercept)        AtBat         Hits        Walks        Years       CAtBat
##  207.8865937   -1.6603271    4.9552022    6.0893338  -12.4996617   -0.2504738
##        CHits        CRuns         CRBI       CWalks    DivisionW      PutOuts
##    0.3954684    1.4147636    1.0607231   -0.7225505 -123.4055931    0.1923965
##      Assists       Errors   NewLeagueN
##    0.5131110   -7.0115407   70.8774554

m.cp3=lm(Salary~Walks+CAtBat+CHits+CRuns+CRBI+CWalks+Division+PutOuts+Assists+Errors+
NewLeague,data=train.data)
m.bic3=lm(Salary~Walks+CAtBat+CHits+CRuns++CRBI+CWalks+Division,data=train.data)
m.adjr3=lm(Salary~AtBat+Hits+Walks+Years+CAtBat++CHits+CRuns+CRBI+CWalks+Division+Put
Outs+Assists+Errors+NewLeague,data=train.data)
```

```
mse.train.cp3=1/210*(sum((train.data$Salary-predict(m.cp3))^2)) #Training MSE for cp
mse.train.bic3=1/210*(sum((train.data$Salary-predict(m.bic3))^2)) #Training MSE for
bic
mse.train.adjr3=1/210*(sum((train.data$Salary-predict(m.adjr3))^2)) #Training MSE for
adjR2

y.hat.test.cp3=predict(m.cp3, newdata=test.data)
y.hat.test.bic3=predict(m.bic3, newdata=test.data)
y.hat.test.adjr3=predict(m.adjr3, newdata=test.data)
mse.test.cp3=mean((test.data$Salary-y.hat.test.cp3)^2)
mse.test.bic3=mean((test.data$Salary-y.hat.test.bic3)^2)
mse.test.adjr3=mean((test.data$Salary-y.hat.test.adjr3)^2)

out.forward=data.frame(Measure=c("Cp","BIC","AdjR2"),Training_MSE=c(mse.train.cp3,mse
.train.bic3,mse.train.adjr3),
                       Test_MSE=c(mse.test.cp3,mse.test.bic3,mse.test.adjr3))
out.forward

##   Measure Training_MSE Test_MSE
## 1      Cp     90888.44 145923.9
## 2     BIC     95658.81 146540.9
## 3   AdjR2     88287.08 134001.7
```

Comment:

The results correspond to models selected using forward selection under three criteria: Cp, BIC, and Adjusted R^2.

Among these, the Adjusted R^2 model has the lowest test mean squared error (134001.7), indicating that it provides the best predictive performance on unseen data. The Cp model has a higher test MSE (145923.9), while the BIC model performs the worst with the highest test MSE (146540.9).

This indicates that, in forward selection, the Adjusted R^2 criterion achieves a better balance between model fit and complexity, leading to improved generalization. The Cp model is slightly less effective, and the BIC model again appears to underfit due to its stronger penalty on including additional variables.

Therefore, for forward selection, the model selected using Adjusted R^2 is preferred for prediction.

6. Repeat 3-4 using backward stepwise selection method.

```
backward.subset=regsubsets(Salary~.,data=train.data,nvmax=19,method="backward")
coef(backward.subset, which.min(summary(backward.subset)$bic)) #Best BIC Model

##  (Intercept)        Walks       CAtBat        CHits       CHmRun    DivisionW
##  149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -129.0165012

coef(backward.subset, which.min(summary(backward.subset)$cp)) #Best cp Model

##  (Intercept)        AtBat         Hits        Walks       CAtBat        CHits
##  151.3709120   -1.5393497    4.6939494    6.1305798   -0.3519984    1.1424820
##       CHmRun        CRuns       CWalks    DivisionW      PutOuts      Assists
```

```
##    2.3082206    0.9269527   -0.5968973 -120.5643234    0.1975004    0.5618678
##       Errors   NewLeagueN
##   -7.1940771   71.8060203
```

```
coef(backward.subset, which.max(summary(backward.subset)$adjr2)) #Best Adjusted R
squared Model
```

```
##  (Intercept)        AtBat         Hits        Walks       CAtBat        CHits
##  151.3709120   -1.5393497    4.6939494    6.1305798   -0.3519984    1.1424820
##       CHmRun        CRuns       CWalks    DivisionW      PutOuts      Assists
##    2.3082206    0.9269527   -0.5968973 -120.5643234    0.1975004    0.5618678
##       Errors   NewLeagueN
##   -7.1940771   71.8060203
```

```
m.cp2=lm(Salary~AtBat+Hits+Walks+CAtBat+CHits+CHmRun+CRuns+CWalks+Division+PutOuts+As
sists+Errors+NewLeague,data=train.data)
m.bic2=lm(Salary~Walks+CAtBat+CHits+CHmRun+Division,data=train.data)
m.adjr2=lm(Salary~AtBat+Hits+Walks+CAtBat+CHits+CHmRun+CRuns+CWalks+Division+PutOuts+
Assists+Errors+NewLeague,data=train.data)

mse.train.cp2=1/210*(sum((train.data$Salary-predict(m.cp2))^2)) #Training MSE for cp
mse.train.bic2=1/210*(sum((train.data$Salary-predict(m.bic2))^2)) #Training MSE for
bic
mse.train.adjr2=1/210*(sum((train.data$Salary-predict(m.adjr2))^2)) #Training MSE for
adjR2

y.hat.test.cp2=predict(m.cp2, newdata=test.data)
y.hat.test.bic2=predict(m.bic2, newdata=test.data)
y.hat.test.adjr2=predict(m.adjr2, newdata=test.data)
mse.test.cp2=mean((test.data$Salary-y.hat.test.cp2)^2)
mse.test.bic2=mean((test.data$Salary-y.hat.test.bic2)^2)
mse.test.adjr2=mean((test.data$Salary-y.hat.test.adjr2)^2)

out.backward=data.frame(Measure=c("Cp","BIC","AdjR2"),Training_MSE=c(mse.train.cp2,ms
e.train.bic2,mse.train.adjr2),
                    Test_MSE=c(mse.test.cp2,mse.test.bic2,mse.test.adjr2))
out.backward
```

```
##   Measure Training_MSE Test_MSE
## 1      Cp     88865.02 134736.7
## 2     BIC     97026.43 153090.5
## 3   AdjR2     88865.02 134736.7
```

Comment:

The results correspond to models selected using backward selection under three criteria: Cp, BIC, and Adjusted R^2.

The Cp and Adjusted R^2 models have identical training MSE (88865.02) and identical test MSE (134736.7), indicating that both criteria have selected the same model. This model provides the best predictive performance among the three.

The BIC model has a higher test MSE (153090.5), suggesting poorer performance on unseen data. This again indicates that BIC may be selecting a model that is too simple, leading to underfitting.

Thus, in backward selection, both Cp and Adjusted R^2 lead to the same optimal model, which is preferred for prediction due to its lower test MSE.

7. Compare the results of the three methods in terms of choice of best model, train MSE and test MSE.

```
final=list(Best_Subset=out.best,Forward_Subset=out.forward,Backward_Subset=ou
t.backward)
final
```

```
## $Best_Subset
##   Measure Training_MSE Test_MSE
## 1      Cp     89790.63 125652.3
## 2     BIC     97026.43 153090.5
## 3   AdjR2     88935.80 130066.9
##
## $Forward_Subset
##   Measure Training_MSE Test_MSE
## 1      Cp     90888.44 145923.9
## 2     BIC     95658.81 146540.9
## 3   AdjR2     88287.08 134001.7
##
## $Backward_Subset
##   Measure Training_MSE Test_MSE
## 1      Cp     88865.02 134736.7
## 2     BIC     97026.43 153090.5
## 3   AdjR2     88865.02 134736.7
```

Comment:

The three subset selection methods—best subset, forward selection, and backward selection—are compared based on their chosen models and corresponding training and test mean squared errors.

In the best subset selection method, the Cp-based model gives the lowest test MSE (125652.3), making it the best among all models in terms of predictive performance. The Adjusted R^2 model performs slightly worse (130066.9), while the BIC model has the highest test MSE (153090.5).

In forward selection, the Adjusted R^2 model performs best with a test MSE of 134001.7, followed by the Cp model (145923.9), while the BIC model again performs the worst (146540.9). Overall, forward selection yields higher test errors compared to best subset selection, indicating comparatively weaker performance.

In backward selection, both Cp and Adjusted R^2 select the same model, resulting in identical training MSE (88865.02) and test MSE (134736.7). This model performs better

than forward selection but is still inferior to the best subset Cp model. The BIC model once again has the highest test MSE (153090.5), suggesting underfitting.

Overall, the best subset selection method provides the best model, particularly under the Cp criterion, as it achieves the lowest test MSE among all approaches. Backward selection performs moderately well, while forward selection is slightly less effective. Across all methods, BIC consistently leads to poorer predictive performance due to selecting overly simple models.